

Boundary Design in AI Systems and User Acceptance: A Case Study of the Facial Recognition Filter <Ectoplasm>

Jang Chaerin

Independent Researcher, mxnsae@gmail.com

Abstract

This paper analyzes how the visual boundary representation of artificial intelligence (AI) systems shapes user experience, using the real-time facial recognition filter <Ectoplasm> as a case study. The title, Ectoplasm, originally refers to the phenomenon in spiritualism where a spiritual entity materializes into a visible form. In this artwork, it is visualized as a "viscous liquid" constraint that restricts the algorithm's search area. Users accept this intentionally implemented boundary by the designer as a rule of play and their own choice. This study explains the mechanism of user illusion occurring in this process by drawing a parallel to the structure of a séance, and examines the politics of AI systems inherent behind technical constraints and the uncritical acceptance of surveillance technology.

1. Introduction

AI systems inevitably determine "what to recognize and what to exclude". While this determination is often regarded as an objective characteristic of the system, it is, in reality, a political choice involving the designer's intention. The real-time facial recognition filter <Ectoplasm> appropriates the occult phenomenon of spiritual entities taking form and materializing in the physical world to visually manifest and reveal the constraints of invisible surveillance systems. By analyzing how users misinterpret and accept the visual constraints presented by <Ectoplasm>, this paper aims to explore humanity's blind faith in modern surveillance technologies and AI systems.

2. Structure and Mechanism of <Ectoplasm>

<Ectoplasm> intentionally utilizes the errors and hallucinations of computer vision systems. The technical structure of the artwork is as follows: (1) The computer vision system generates a "hallucination," forcefully finding traces of faces within noise regardless of the

actual presence of a human face. (2) As if spiritual energy were materializing, a viscous liquid visualization drips down the screen, carving out specific zones where the algorithm is "permitted" to search for human traces. (3) Facial recognition bounding boxes are generated exclusively within the area inside this liquid boundary. The crux here does not lie in the technical objectivity of the system accurately capturing a human, but in the fact that the illusion (facial recognition result) arbitrarily created by the machine in empty space is materialized as a bounding box only within the liquid boundary permitted by the designer.

3. Boundary Design and the Mechanism of Illusion

The relationship between the liquid's form and the bounding boxes is strictly under the intentional control of the designer. However, at the level of user experience, this constraint is translated into an entirely different meaning.

- The Reality of Design: The designer arbitrarily

restricts and excludes the recognition area through the liquid visualization.

- The User's Experience: The user perceives the designed constraint as a tool for play and actively controls their body to obtain a bounding box.

Users in the initial stage, unaware of the system's hidden rules, observe that their face outside the liquid boundary fails to form a bounding box and mistakenly believe that "the system failed to recognize me". Faces that do not enter the liquid boundary are thoroughly excluded, yet users misinterpret this as the natural disposition of the system. Subsequently, users voluntarily move to align their bodies within the boundary to be included as an object of recognition. Oppressive control and exclusion are transformed into the user's voluntary play, ultimately demonstrating how technological constraints induce a "bodily alignment" that forces the human body to conform to algorithmic standards.

4. The Occultism of Surveillance Technology and the Politics of AI

<Ectoplasm> is structurally identical to the 19th-century séance in that it induces hallucinations, making what does not exist appear as if it does. As media scholar Jeffrey Sconce points out, historically, new technological media have always been perceived as a means to materialize invisible spiritual entities [1]. Just as a séance manifested formless beings in the physical world through ectoplasm, computer vision also conjures illusions and creates algorithmic hallucinations. Much like Hito Steyerl criticized the apophenia of systems that forcefully read patterns into meaningless data [2], the computer vision in this artwork compels us to believe in an impossible sight as truth. Every AI system must determine what to receive as input, what to process, and what to return as a result. These determinations are not

mere technical limitations but the political choices of the designer regarding who sets these boundaries [3]. Nevertheless, we believe the arbitrary and false visions of the system, materialized on the screen, to be real. The convergence point of surveillance technology and the occult lies precisely here: in our willingness to accept the system's fabricated visions as truth.

5. Conclusion

Through <Ectoplasm>, we can observe the process by which the politics of AI design is translated into user experience through visual metaphors. Users accept the designer's choices as the inevitable nature of the system, embracing the constraints of the surveillance system as their own elective play. The parallelism between surveillance technology and séances—both defining the invisible and materializing virtual entities before our eyes—poses a fundamental question about how we believe in and become subordinated to technology. Critically understanding the mechanisms of boundary design and the operation of power is a crucial task for future AI system research.

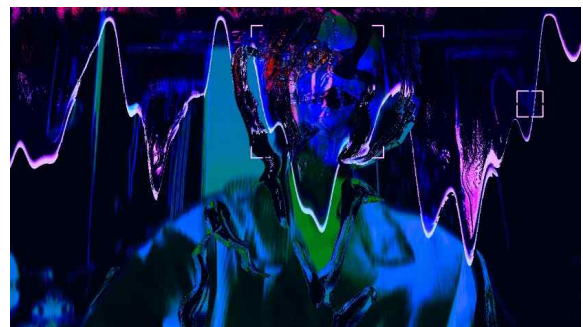


Figure 1. Execution screen of the real-time facial recognition filter <Ectoplasm>

References

- [1] Sconce, J. (2000). *Haunted Media*. Duke Univ. Press.
- [2] Steyerl, H. (2016). "A Sea of Data". *e-flux journal*.
- [3] Crawford, K. (2021). *Atlas of AI*. Yale Univ. Press.